Practical 3: An introduction to the rock-forming minerals

On successful completion of this practical, you should be able to:

- Identify common rock-forming minerals in hand specimen and thin section
- Describe rock-forming minerals in plane-polarized and cross-polarized light
- Estimate the modal mineralogy of an igneous rock

You will also practice the following skills:

- Describe the mineralogical and optical properties of common igneous minerals, including: relief, colour, pleochroism, cleavage, twinning, extinction angle.
- Create accurate sketches of important igneous textures and mineralogical features
- Interpret the order of crystallization using observations of igneous textures

The Practical Reference Handout

This handout contains a series of useful diagrams and notes that will assist you to make systematic descriptions and interpretations of igneous rocks. You will be allowed to bring your own annotated copy to the practical test in December, so **it is a good idea to add your own annotations as you go along!**

Some useful pages for this practical:

- Colour, crystallinity and grain size (p 2)
- Table of mineral optical properties and distinguishing features (inside back cover). *Bear in mind that minerals do not always show the ideal features listed, and may show other useful features that are not listed.*
- Detailed notes on mineral properties (pp 22-24)
- Distinguishing between clinopyroxene and orthopyroxene (p 25)
- Chart for estimating modal proportions (p 28)
- Terms for describing crystal shapes (p 15-16)

Part 1 (15 minutes): The rock-forming minerals in hand specimen

A selection of hand specimens of the rock-forming minerals are laid out on the benches. Take a look at the different specimens and make notes on their characteristic features, such as:

- Habit
- Colour
- Lustre (dull, shiny, metallic, etc)
- Any visible cleavage or twinning
- Anything else that will help you recognise this mineral if and when you see it again

You may wish to take photographs to go alongside your notes. Remember to include a scale.

Part 2 (90 minutes): Systematic characterization of igneous rocks in thin section

There are 6 thin sections to look at in this practical. You can look at them in any order.

The aim of this class is to practice making systematic observations and descriptions of the mineralogy and texture of these rocks. You will probably not have time to work on each thin section in detail, but do aim to have looked at each one before the end of the practical.

I have provided a classification and a short description for each thin section. Check that you can identify all the features mentioned. For each thin section there are some questions to guide your petrographic observations. Pay particular attention to the characteristic features of the main silicate minerals. By the end of the practical you should have collected enough observations and notes that you will be able to recognise these minerals and their optical properties when you meet them again in other thin sections.

Terminology	
Modal mineralogy	The percentage of the different minerals that make up the rock, totalling 100%. It is possible to quantify the modal mineralogy using a technique called point counting, but this is extremely time-consuming so you will estimate the modal mineralogy by eye.
Crystal shapes	Different minerals have different characteristic shapes. For example, plagioclase tends to form tabular (blocky rectangular) crystals. The shape also depends on whether the crystal faces are well-developed, moderately developed, or absent. You will find some useful terms on p 15-16 of the Practical Reference Handout, and we will discuss crystal shapes in more detail next week.

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This rock is a **biotite hornblende granite**. The **hornblende** has a bluish-green colour in some orientations. Its characteristic lath shape and diamond cross-section are well developed. The rock contains **biotite**, which in some thin sections is partly altered to **chlorite**. The rock also contains **quartz, plagioclase** and **alkali feldspar**. The **plagioclase** is zoned, and is partly altered to sericite, a muscovite-like white mica: colourless, with moderate birefringence and mottled extinction. Some thin sections also have accessory **titanite** (also called sphene).

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral. Describe the characteristic crystal shapes of the different minerals.

Plagioclase	~25%	1-6 mm	tabular to subhedral
Alkali feldspar	~35%	0.5-3 mm	tabular to subhedral
Quartz	~20%	0.2-3 mm	anhedral; partially recrystallized
Biotite	~5%	1 mm	subhedral laths
Hornblende	~14%	0.5-1.5 mm	euhedral; elongate laths
Titanite	1%	<1 mm	euhedral
Oxide	trace	<1 mm	subhedral to anhedral

Estimate the relative proportions of plagioclase and alkali feldspar.

I found more alkali feldspar than plagioclase, but the modal proportions vary between thin sections.

Describe the colour of the biotite, and sketch the relationship between biotite and chlorite. What process could cause this texture to form? Discuss you answer with a demonstrator.

The biotite is pleochroic in brown to straw yellow, but the brown colour has a greenish tint to it. The green chlorite is replacing the biotite – best seen at the edges of the crystal and along the cleavage traces. This is caused by hydrous alteration.

81108

This rock has a very unusual composition and mineralogy. It is a **hornblende aegirine alkali syenite**. It contains an **alkali feldspar** micro-perthite. The rock also contains **hornblende** and an unusual variety of sodium-rich clinopyroxene: a pale green **aegirine-augite**. You may see accessory **titanite** in this rock. *(The mineralogy of the thin section marked 81108X is somewhat different to that given above, but it has the best examples of titanite.)*

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral. Describe the characteristic crystal shapes of the different minerals.

Alkali feldspar	85%	1-3 mm	tabular to anhedral
Aegirine augite	up to 10%	0.2-2 mm	subhedral
Hornblende	~5%	0.2-1 mm	subhedral
Titanite	<1 %	0.5-2 mm	subhedral to anhedral
Oxide	<1 %		

81108X contains plagioclase and quartz in addition to alkali feldspar

The alkali feldspar in this rock has a **perthitic** texture, which is created when the alkali feldspar solid solution exsolves (separates out) into sodium-rich and potassium-rich lamellae within the mineral. In this thin section, the exsolution lamellae are well-developed. Check that you can identify this texture and make a labelled sketch of what you observe. What kind of alkali feldspar is this: sanidine, orthoclase or microcline? What does the internal texture tell you about the cooling rate of this igneous rock?

This is orthoclase: it has simple twinning and perthitic exsolution. All the crystals have distinctive blebs of Na-rich albite, and some of these blebs have lamellar twins. The exsolved lamellae are quite coarse, which suggests that the rock must have cooled slowly. But the cooling rate was not slow enough to cause Al:Si ordering in the alkali feldspar – otherwise we would have microcline.

In 81108X some of the feldspars show concentric zonation. The plagioclases are altered to sericite, and the alkali feldspars have perthitic exsolution.

Examine the relationship between the hornblende and the aegirine-augite. Some of the hornblende forms rims around the aegirine crystals. Why might these two minerals have this textural relationship (ask a demonstrator)?

The hornblende must have crystallized later than the augite. Pyroxenes are dry, whereas amphiboles can take water into their crystal lattices. Progressive crystallization of anhydrous minerals caused the remaining silicate liquid to become more hydrous, allowing hornblende to start crystallizing. Hornblende has a very similar crystallographic structure to pyroxenes, so it uses the clinopyroxene as a convenient nucleation template.

R1 *Rum layered intrusion, Inner Hebrides, Scotland*

This rock is an **olivine dolerite** (or olivine gabbro). It consists of **olivine**, **augite**, **plagioclase**, and an **opaque oxide**. The olivine has high relief and curving fractures. The sample is quite fresh, but some of the olivines show characteristic alteration to serpentine around the crystal rims and inside the fractures. The interference colours of the olivines range from low up to at least second-order yellow, depending on the orientation of the crystal in the thin section. Most of the olivines in this rock are subhedral or anhedral (i.e. they have

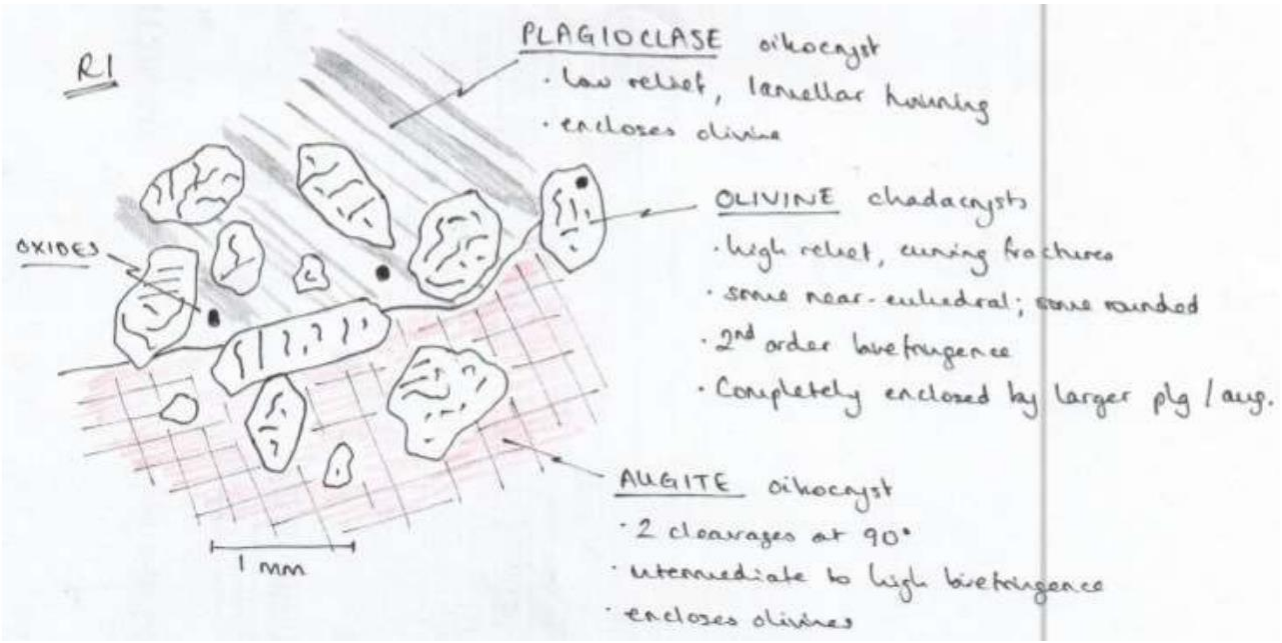
rounded shapes with few well-developed crystal faces), so it is unlikely that you will be able to measure extinction angle.

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral, and classify the rock as fine, medium or coarse-grained *(use the Reference Handout)*. Describe the characteristic crystal shapes of the olivines and the oxides.

Olivine	~40%	0.5-1 mm	subhedral to anhedral; some are euhedral
Plagioclase	~30%	1-2 mm	anhedral
Augite	~30%	<1 mm up to 4 mm	anhedral
Oxide	~1%	~10 um	euhedral; cubic habit

The modal abundance of plag and cpx varies between thin sections. This is necessarily an estimate, so give yourself a ±5-10% margin of uncertainty.

Sketch the texture of this rock. Look at p 20 of the Reference Handout and see if you can find a name for this texture. What can it tell you about the order of crystallization? Discuss your interpretation with a demonstrator.



Olivines are enclosed by large anhedral crystals of plagioclase and/or augite (cpx). This is a **poikilitic** texture. The olivine crystals formed first, and were then enclosed by later-growing plagioclase and augite. The oxides are usually enclosed within plagioclase and augite, but there are some oxide inclusions within the olivines too. There must have been small amounts of oxide crystallizing at the same time as all the other minerals.

We will discuss textures, and order of crystallization, in much more detail next week.

RFM7

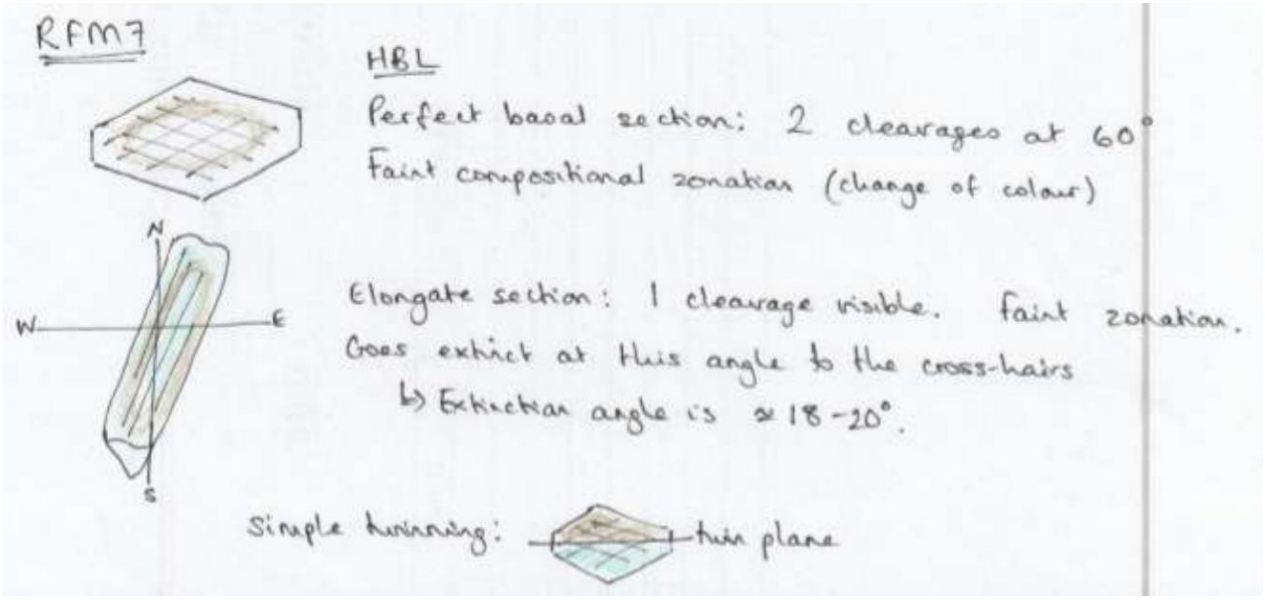
This rock is a **hornblende syenite**. It contains **alkali feldspar**, **plagioclase** and **hornblende**. Plagioclase occurs as large crystals showing characteristic lamellar twinning.

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral. Describe the characteristic crystal shapes of the different minerals.

Plagioclase	35%	0.5-1.5 mm	tabular to subhedral
Alkali feldspar	40%	0.5-3 mm	tabular to subhedral
Hornblende	25%	0.5-1 mm	euhedral to subhedral
Apatite	trace	<0.1 mm	euhedral prismatic inclusions in hornblende

Make sketches of the hornblende crystals. Try to find (a) crystals where both cleavage planes are visible, and (b) crystals where only a single cleavage plane is visible. Label your sketches with the orientations of the refractive indices and their associated pleochroic colours. What is the extinction angle in the one-cleavage sections? What other notable features can you observe in the hornblende?

Some hornblendes contain small inclusions of the mineral **apatite**, a phosphate mineral. Apatite has a hexagonal prismatic habit. In PPL it is colourless and has high relief. In XPL, it has low first-order interference colours. Make sure you can identify the apatite.



Basal sections through hornblende have the characteristic diamond-like shape, two cleavages at ~60°, and symmetrical extinction bisecting the cleavage traces. In the one-cleavage section, the extinction angle is about 18°. The hornblende in this thin section shows some simple twinning. The pleochroism is strong, in shades of yellow, brown and green. Some crystals appear to have concentric rings picked out by different shades of brown or green. This is compositional zoning.

SW

This rock is a **biotite muscovite granite**. It contains **biotite, muscovite**, **plagioclase, alkali feldspar** and **quartz**. The alkali feldspar is a micro-perthite. The host mineral is microcline (note the faint cross-hatched twinning seen in some grains). Some of the exsolved Na-rich alkali feldspar shows lamellar twinning. That is because it is albite, which as well as being an alkali feldspar, is one of the endmembers of the plagioclase series. Some of the K-feldspars show simple twinning.

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral. Describe the characteristic crystal shapes of the different minerals.

Plagioclase	15%	0.5-1 mm	tabular
Alkali feldspar	25%	0.5-1 and 4-5 mm	tabular to subhedral

Quartz	40%	<0.1 and 1-2 mm	anhedral
Muscovite	~8%	0.5-1.5 mm	laths
Biotite	~7%	0.5-1 mm	euhedral laths

Make notes on the optical characteristics of the biotite: pleochroism, cleavage, appearance close to extinction. Notice the small, dark circles inside the biotite crystals. These are called **pleochroic haloes**. Ask a demonstrator how the pleochroic haloes are formed.

Biotite has two pleochroic colours: brown and straw yellow. It has one perfect cleavage. It has straight extinction, and a speckled appearance when close to extinction. The interference colour is high third-order. The pleochroic haloes form around tiny inclusions of the mineral **zircon** (the zircons themselves are mostly too small to see). Zircon can accommodate a tiny amount of uranium in its crystal structure. When the uranium undergoes radioactive decay, it emits alpha particles that cause radiation damage to the surrounding biotite lattice. This creates the pleochroic haloes.

Some of the biotite, muscovite and quartz crystals have undulose extinction. Quartz is often present as small interlocking grains, which is evidence of recrystallization. What process could have caused these features?

Undulose extinction and recrystallization are evidence that this rock has undergone some minor deformation.

The plagioclase crystals often have a dusty appearance: they have been hydrothermally altered to a clay mineral called **sericite**. What other features do you observe in the plagioclase? Make a labelled sketch to illustrate them.

Plagioclase has multiple lamellar twins and, in this thin section, it is compositionally zoned. In this section, the sericite is usually confined to the crystal cores.

DWR10

This rock contains an opaque oxide. This is **spinel**. What are the two silicate minerals?

The silicate minerals are plagioclase and orthopyroxene. Opx is identified by its low first-order interference colours, and straight extinction in elongate crystals and one-cleavage sections. Some of the opx crytals have brightly birefringent exsolution lamellae of clinopyroxene. There are also some small clinopyroxene crystals in some of the thin sections.

Estimate the modal mineralogy of this rock. Estimate the range of crystal sizes for each mineral. Describe the characteristic crystal shapes of the different minerals.

Orthopyroxene	~40%	0.5-2 mm	euhedral, prismatic
Plagioclase	~40%	1-4 mm	subhedral
Spinel	~20%	0.1 mm	euhedral, cubic

What can you learn about the order of crystallization by looking at the texture of this rock?

The texture is poikilitic: opx and spinel chadacrysts are enclosed in large oikocrysts of plagioclase. Some of the spinels are enclosed within larger opx crystals. These textural relationships suggest that spinel was the first mineral to start crystallizing; then spinel and opx crystallized simultaneously. Plagioclase must have started crystallizing later, because it fills up the interstitial spaces between the other minerals.

Consolidate your observations

- Quartz, alkali feldspar and plagioclase have fairly similar optical properties, but there are several features that can be used to tell the difference between them. What techniques or observations are most useful for identifying and distinguishing these three minerals?

Quartz, alkali feldspar and plagioclase are all low relief, colourless, and have low first-order interference colours. However, there are several features you can use to tell the difference between them. You should come up with your own list of things to look for. Here is mine:

	PPL	XPL	Shape
Quartz	No cleavage. Never shows cloudy alteration.	No twinning. No exsolution. May show undulose or patchy extinction if deformed	Usually rounded
Plagioclase	Cleavage rarely visible. May show cloudy alteration to sericite.	Multiple lamellar twinning. Never shows exsolution.	Usually tabular or acicular
Alkali feldspar	Cleavage rarely visible. May show cloudy alteration to sericite.	In a volcanic rock: usually sanidine. Simple twinning; no exsolution. Use twinning and cleavage to distinguish from quartz. In a plutonic rock: usually orthoclase or microcline. Both can show perthite exsolution lamellae. Orthoclase: simple twinning. Microcline: cross-hatch twinning.	Usually tabular